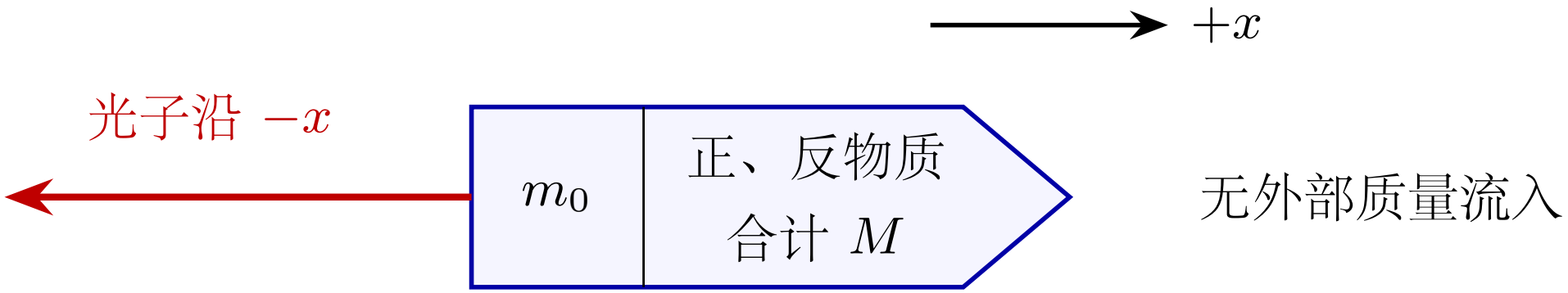
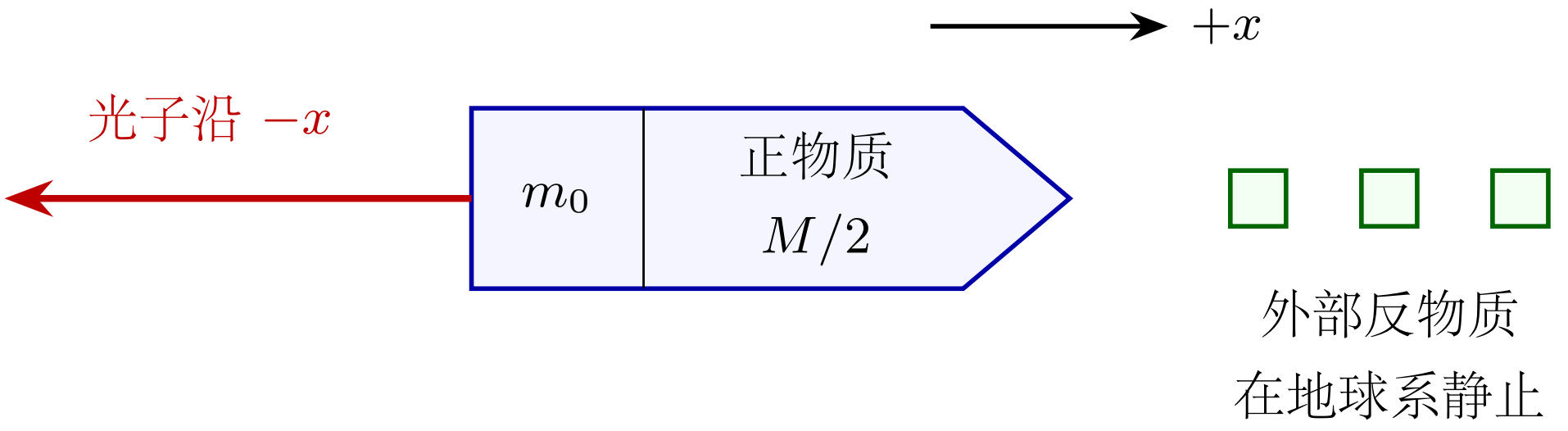
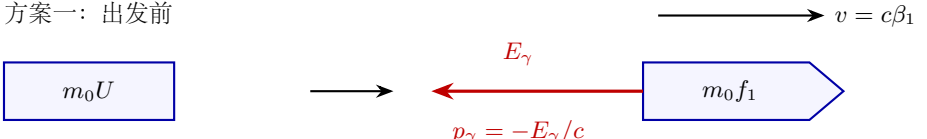
方案一：初始船载 $m_0 + M$



方案二：初始船载 $m_0 + M/2$



方案一：出发前



$v = 0, \quad p_{\text{initial}} = 0$

初始总能量 m_0Uc^2

$E_{\text{ship}} = \gamma_1 m_0 f_1 c^2$

$p_{\text{ship}} = \gamma_1 m_0 f_1 \beta_1 c$

方案二：船与待收集部分

$v_{\text{in}} = 0$

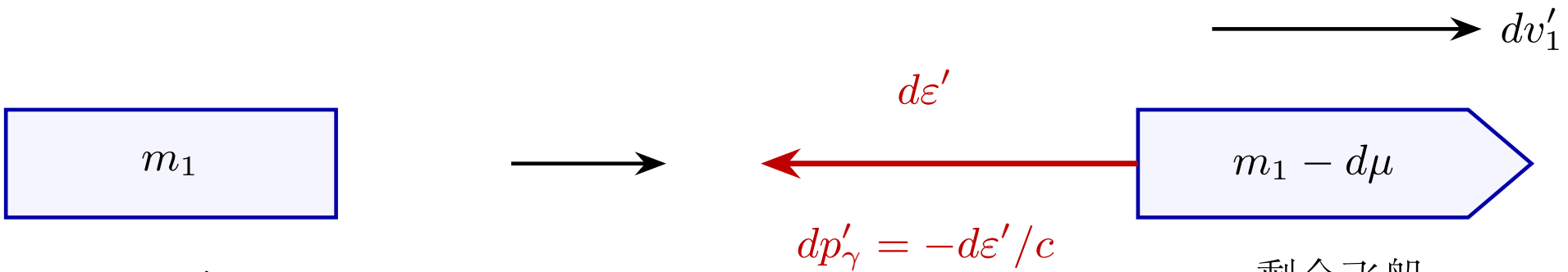
两者初始动量均为零

总能量 $m_0(A + X)c^2$

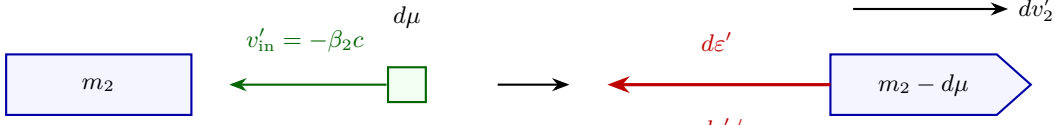
$E_{\text{ship}} = \gamma_2 m_0 f_2 c^2$

$p_{\text{ship}} = \gamma_2 m_0 f_2 \beta_2 c$

瞬时静止系 S'，船内消耗 $d\mu = \alpha\, d\tau$



瞬时静止系 S'：入射反物质与船内等质量正物质湮灭



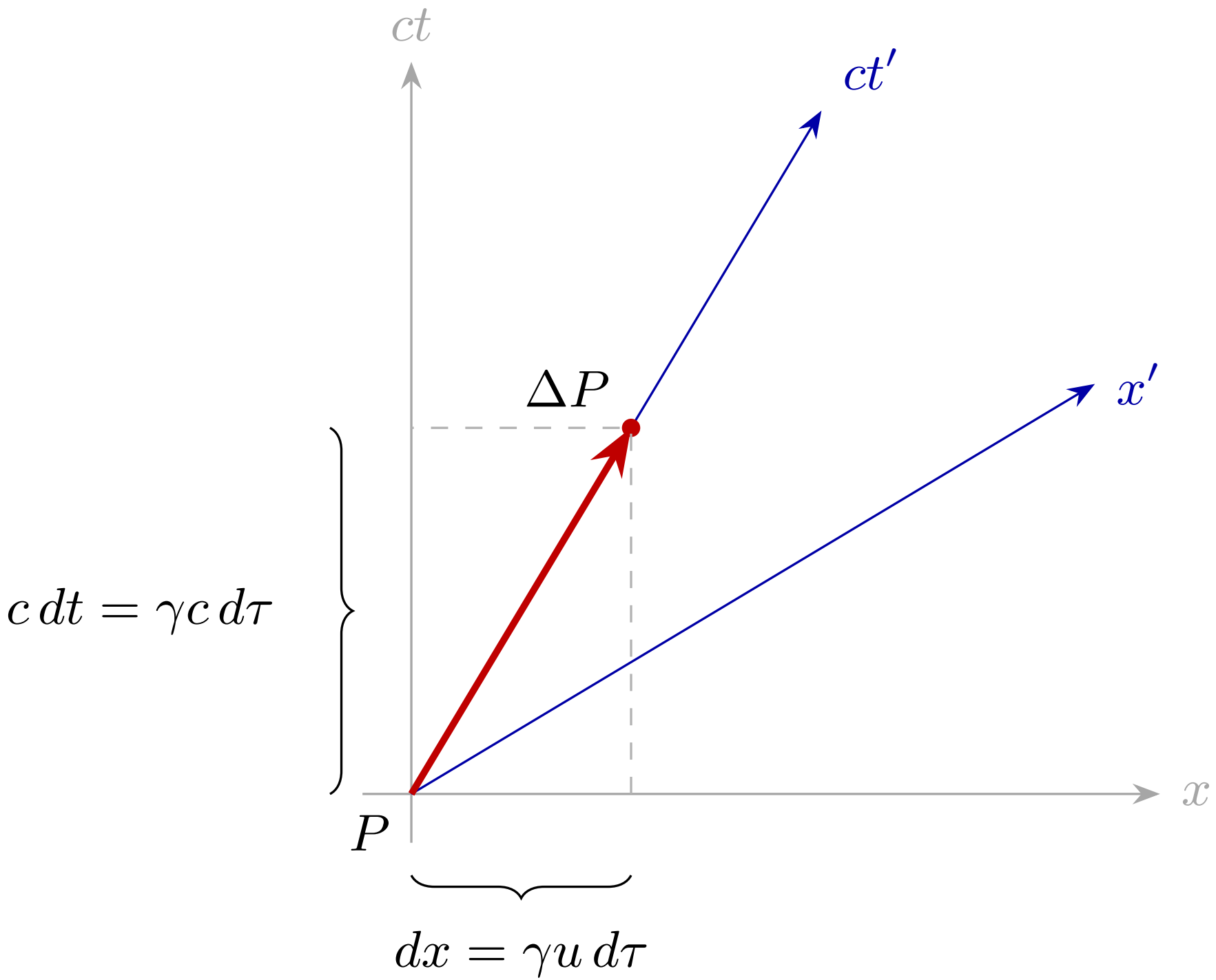
过程开始时 $v' = 0$

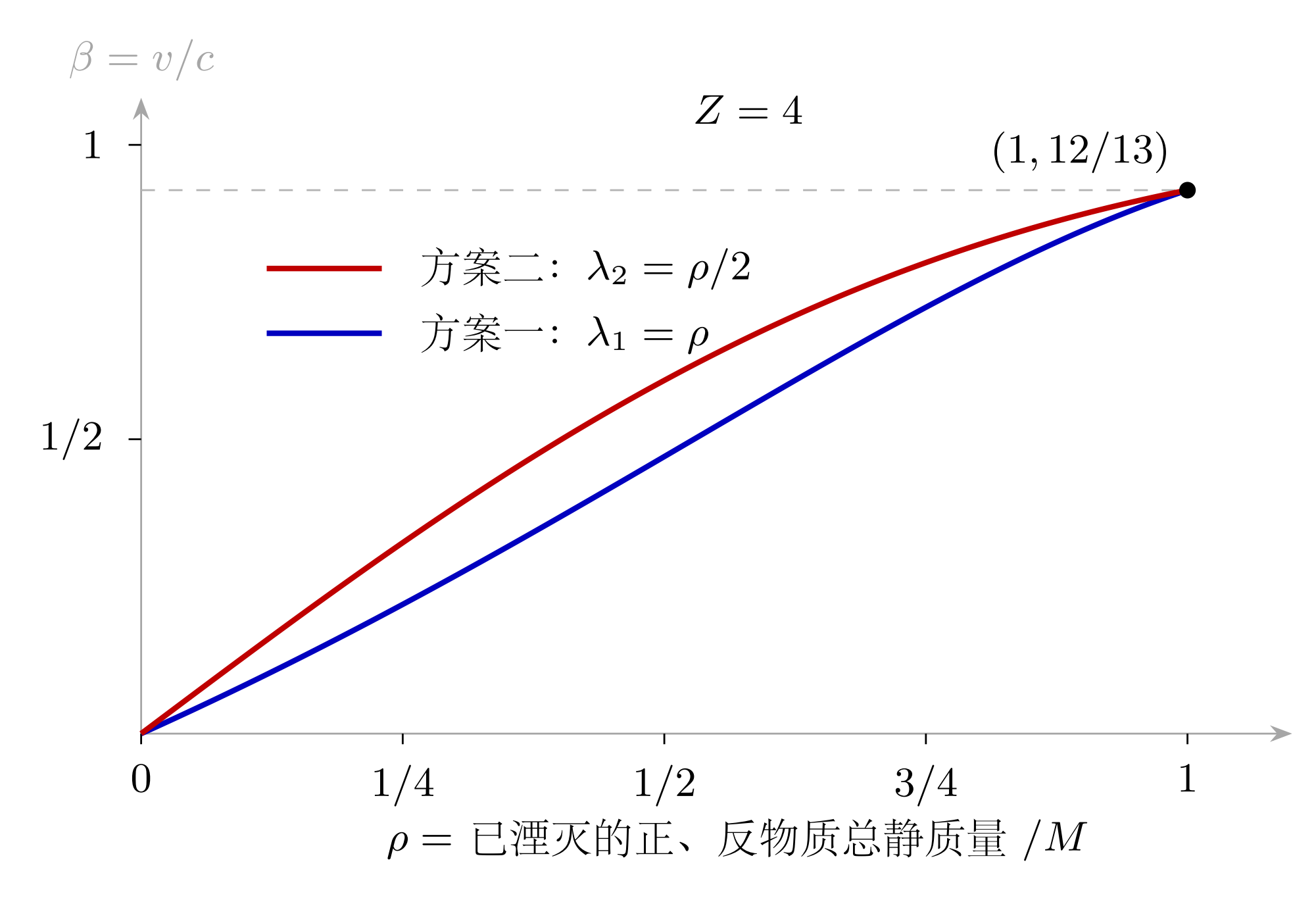
$dE'_{\rm in} = \gamma_2 d\mu c^2$

$dp'_{\rm in} = -\gamma_2\beta_2 d\mu c$

末态飞船

$p' = m_2\, dv'_2$（一阶）


ct
ct'
ΔP
x'
$c\,dt = \gamma c\,d\tau$
x
P
$dx = \gamma u\,d\tau$


$\beta = v/c$
$Z = 4$
1
$(1, 12/13)$
方案二：$\lambda_2 = \rho/2$
方案一：$\lambda_1 = \rho$
1/2
0
1/4
1/2
3/4
1
$\rho =$ 已湮灭的正、反物质总静质量 $/M$